COMP1801 - Machine Learning Coursework Report

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1. Executive Summary

This report examines the challenges that the company is facing to predict the durability of the metal parts based on their Lifespan, as well as specifying that the 1500-thershold is an important component in classifying which parts are usable. This process will be incredibly helpful as it would assist in managing quality assurance and improve the production process which would lead to an early identification of all defective metal parts, therefore reducing waster and improving the production efficiency.

This problem is solved by using machine learning models by making use of Linear Regression and Artificial Neural Network (ANNs) to predict the continuous Lifespan of the metal. Furthermore, the Logistic Regression and Artificial Neural Network were implemented to classify whether a metal part were useable or not.

During model testing and hyperparameter optimization, the ANN model emerges as the most successful for both tasks, surpassing both the Linear Regression and Logistic Regression. The ANN model attained a notable accuracy of 92% in predicting the Lifespan and 93% in classifying the metal parts. In light of these finding, it is advisable to implement the ANN model for assessing the durability of the metal parts, as it is able to provide higher accuracy and management of feature complexities, which will help deliver valuable insights for the organization's quality control measures.

2. Data Exploration

In order to investigate the Lifespan of the metal parts, the dataset was imported into a DataFrame through the pd.read_CSV() function. The structure was examined by data.head().The dataset encompasses production attributes, chemical compositions, defect tallies and the target variable the Lifespan signifying the durability of the metal parts.

In order to gain a more comprehensive understanding of the variability present within the dataset, a summary statistic was computed for the target variable alongside key variables. As the Lifespan variable assesses the durability of the metal parts, it shows a mean value of 1298 and standard deviation of 340 which indicates a substantial range within the data. The recorded minimum and maximum values were 418 and 2135, respectively, suggesting tat a diverse array of conditions may exert a significant influence on durability.

The histogram illustrated a moderate skewed distribution with most of the values clustered around the mean and a limited outliers at both ends. These outliers can signify instances where the metal parts have incredibly high or low durability which can potentially reflect on the different combinations of chemical compositions or production attributes.

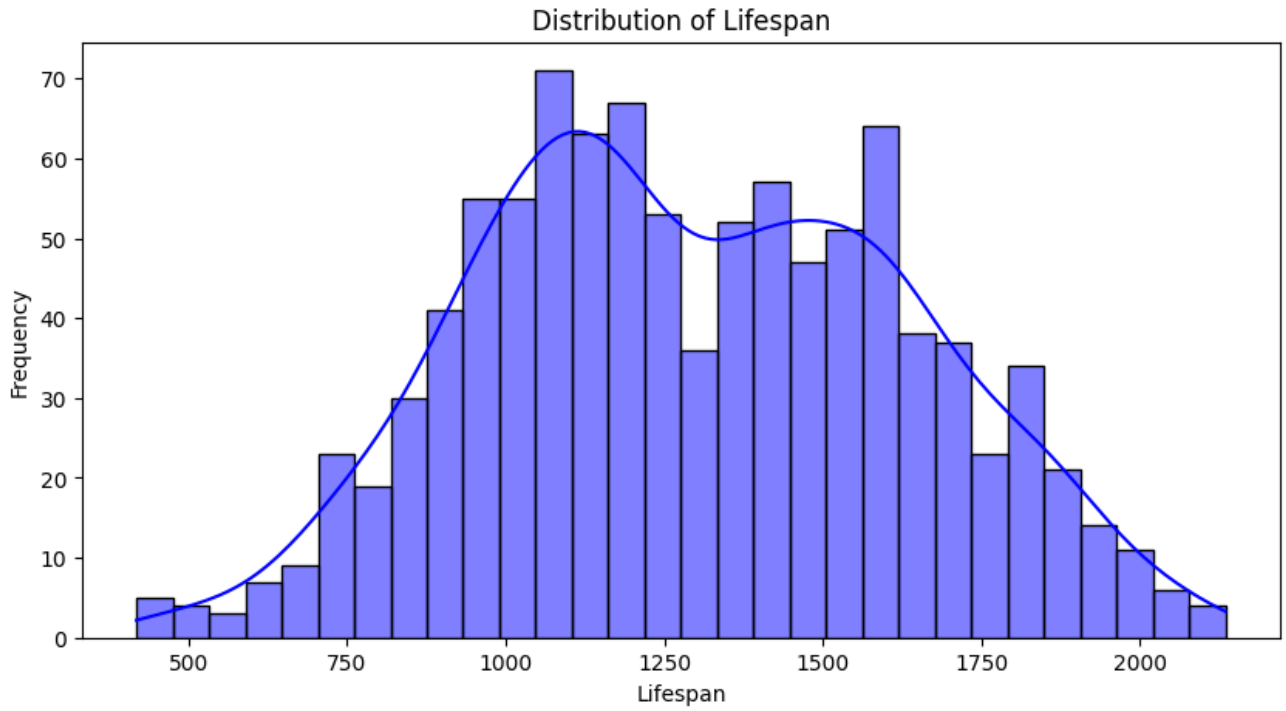


Figure 1 Histogram of Lifespan

The correlation heatmap shows a complete picture of how the variable are related to each other. Nickel% exhibits a mean value of 60.4 and a moderate positive correlation with Lifespan of 0.32, which suggests that a higher nickel concentration may enhance its durability. In comparison, smallDefects averages at 17 with a weak correlation to Lifespan of 0.10 indicating that while defects can affect durability, their relationship with Lifespan is not strongly linear. Iron% averages at 25.3, showing a negative correlation of -0.25 with Lifespan which signifies a detrimental effect on the metal parts. Furthermore, the production attributes such as coolingRate, HeatTreatTime and quenchTime which show a weak correlation with the target variable; however, they can possibly indirectly affect the durability due to the quality of the metal through their chemical compositions.

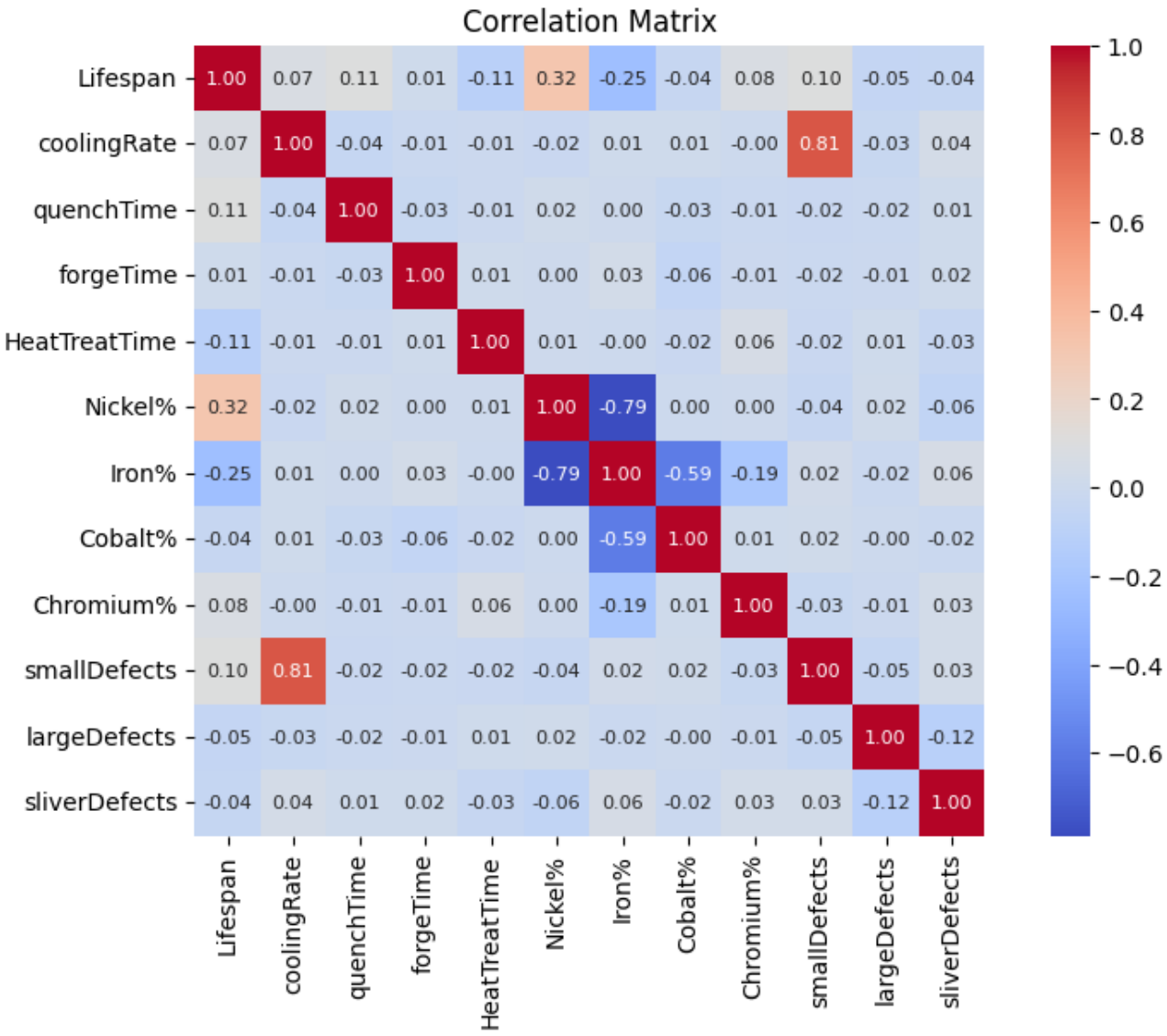


Figure 2 Correlation Heatmap

The boxplot further validates these observations, for instance, Nickel% shows a consistent distribution with negligible outliers which will help enhance the accuracy as a predictor. The smallDetects reveal many outliers which suggest occasional incidences of remarkably high defect frequencies which are likely to contribute to premature failures.

From this analysis several import patterns were revealed. The chemical composition of the materials was identified as a crucial factor in determining the durability , with Nickel% having a positive impact on the Lifespan, whereas Iron% having the opposite. Although, both Chromium% and Cobalt% has less correlations, however they still can influence basic alloy characteristics. The metrics related to defects, while playing a small role, it can still help understand any flaws in the manufacturing process. The production attributes display a weak correlation; however, they likely affect the Lifespan indirectly by impacting the microstructure of the metal parts.

A regression methodology is the most appropriate framework for estimating the Lifespan due to its continuous nature. For this objective, two distinct models have been selected: Linear Regression and an Artificial Neural Network (ANN). (Hope, 2020) explains that the linear regression model is best used when there is a strong linear interpretability between the labels and the features like Nickel% and Iron% show a moderate linear correlation with Lifespan. (Krenker et al., 2011) describes the ANN regression model that is robust and

capable of capturing complete dependencies like the coolingRate, HeatTreatTime and quenchTime, which will help uncover all the hidden non-linear relationships.

3. Regression Implementation

3.1 Methodology

(Uyanık & Güler, 2013) defines linear regression as one of the most popular machine learning models due to its simplicity and interpretability, when focusing on analysing the relationship between dependent variables and independent variables are linear. Considering that certain features within the dataset display a linear correlation with the label , this model emerges as an obvious choice regarding the magnitude and patterns of the correlations between the independent and dependent variables via its coefficients.

ANNs was selected for its ability to represent the complicated non-linear correlations in the dataset. The Lifespan may have been affected by complex interactions among various features which the linear models may fail to capture. (Lecun et al., 2015) explains that ANNs can capture these complex relationships by their multiple layers of interconnected neurons which will enhance the accuracy. The ANNs becomes a powerful tool when adapting to diverse feature types (Numerical and Categorical).

As the preprocessing routine is important to ensure the models operate at the highest efficiency. The Categorical feature was encoded to implement into numerical variables which can be processed by the models. To do this, One Hot Encoding was used as it effectively prevents any ordinal relationship among the categorical values (Poslavskaya & Korolev, 2023). The Numerical features were scaled via StandardScaler to maintain uniform scale among all features, as these models are sensitive to scale, which is noticed in (Aldi et al., 2023) research giving a better result when using StandardScaler. The dataset has been divided into training, validation and test following an 80-10-10 split. The 80% will be used to train the model, another 10% is used for validation to tune the hyperparameters and the rest 10% is used to test for a conclusive assessment. This strategy will help to mitigate the risk of overfitting and ensure the model adapts accordingly to any new data.

For tuning the hyperparameter for Linear Regression, the polynomial degrees ranging from 1-4 were experimented to capture any non-linear correlations. This will be immensely helpful as it will allow the model to fit any curved relationship present within the dataset (Stimson et al., 1978). The hyperparameter tuning for the ANNs will include the quantity of the hidden layers and the number of neurons that are allocated to each layer. Furthermore, activation functionalities like ReLu and tanh will be used as it defines the output of each neuron, impacting the network's ability to learn complete links, which is explained by (Sharma et al., 2020). Finally, the learning rate will be adjusted to determine the speed at which the model converges throughout the training process.

3.2 Evaluation

For the Linear Regression model, experimentation of the polynomial degrees ranging from 1 to 4 to capture the non-linear relationships in the dataset which was explained in part 3.1. This experimentation was done on two different datasets where one dataset included the relevant numerical and categorial features, where the categorical features were one hot encoded, and the other dataset was only the numerical features.

Table 1

Validation Results For Polynomial Regression (One Hot Encoding)			
Degree	Validation MAE	Validation MSE	Validation R^2
1	263.6	98599.5	0.12
2	143.7	29607.3	0.74
3	251.4	100978.2	0.10
4	214.3	99169	0.11

Table 2

Validation Results For Polynomial Regression (Numerical Features)			
Degree	Validation MAE	Validation MSE	Validation R^2
1	269.8	106711.0	0.05
2	158.0	37482.7	0.67
3	171.1	43526.2	0.61
4	164.1	40837.6	0.64

Table 3

Metric	Polynomial Degree 2
Mean Absolute Error (MAE)	112.0
Mean Squared Error (MSE)	19054.6
R^2	0.80

After the best model was evaluated with the test set the MAE represents the average size of the model's inaccuracies. An MAE of 112.0 indicates the results of the Linear Regression model's predictions differed from actual lifespan values by approximately 112 units on average. Although this is an acceptable inaccuracy considering it is a simple model, it points out that the model may not be particularly precise. MSE penalise greater errors more severely than MAE. This rather high score indicates that the model produced a few substantial errors in forecasting the Lifespan, affecting its performance. The MSE is highly susceptible to outliers, which implies that substantial departures from actual values have an imbalance on the outcome. The R^2 value of 0.80 means that the model accounted for 80% of the variation in the test set.

For the ANNs model, the optimization of the several hyperparameters include the number of hidden layers, neurons per layer, activation functions and the learning rate which was explained in part 3.1. The dataset used to train, validate and test the model was similar to the Linear regression.

The hidden layer size hyperparameter was experimented on 3 sizes (50,), (100,), (100,50), this was done to understand the different network complexities. The single layer network complexities (50,) and (100,) may underfit or overfit, while the multi-layer network complexities (100,50) gain the ability to capture more complex patterns, to balance the model's accuracy and generalization. For activation, the tanh and ReLu functions were used because tanh helps with the non-linear relationship as well as preventing larger gradients, while ReLu will help to promote faster merging in deeper networks. For learning rate, 0.001, 0.01 and 1 were used because a higher rate would cause the model to overshoot while a lower one would slow the training process down. For the max iter, 250 and 500 were used because it is necessary for the model to have enough training time, where 250 iterations were used for a faster learning time and 500 iterations for a slower iteration time.

Table 4

Best Hyperparameter Of ANNs Model		
Hyperparameters	One Hot Encoding	Numerical Features
Hidden Layer Size	(100,)	(100,)
Activation	tanh	tanh
Learning Rate	0.1	0.1
Max Iter	500	500

Table 5

Metric	One Hot Encoding	Numerical Features
Mean Absolute Error (MAE)	70.5	85.46

Mean Squared Error (MSE)	7774.4	11560.0
R^2	0.92	0.88

Table 6

Metric	Linear Regression	ANNs
Mean Absolute Error (MAE)	112.0	70.5
Mean Squared Error (MSE)	19054.6	7774.4
R^2	0.80	0.92

By comparing the performance for both the models, it is clear hat the ANN performs much better than the linear regression across all the metrics. ANN has a MAE of only 70.5, which shows that it is much more accurate than those of the Linear Regression. The MSE is 7774.4 suggesting that the model has made fewer large error when prediction the lifespan, suggesting that ANN has an overall better accuracy and variation in its prediction. ANN has a R^2 of 0.92 which is significantly better than that of linear regression, which shows that the ANN is better at capturing the underlying patterns and relationship between the features and the label.

3.3 Critical Review

Linear regression had provided a strong baseline for understanding the model capacity to learn, however the ANN model was much better at capturing the complex, non-linear relationship that are present in the data as shown by a better R^2 and lower MAE and MSE. In the preprocessing routine, experimenting on the one hot encoding for categorical features and standard scaling for numerical features, further improved the learning capacity of the model.

Even though the ANN performed well, it still had many areas of improvement as it is easily prone to overfitting due to it complex nature. This could have been solved by using different regularization techniques such as early stopping or dropout. Exploring deeper networks like the convolution neural networks (CNNs) would have had improved the capability to capture the complex and hidden data patterns. XGBoost model might have been better in this scenario as it is able to manage complex relationship without overfitting.

4. Classification Implementation

4.1 Feature Crafting

The objective in this section is to build a binary classification on a 1500-hour threshold, which allows to differentiate metal parts to be defective or not in relation to their Lifespan. Nonetheless, this might result in an oversimplified situation and can perhaps disregard patterns and insights that would be identified when the data is divided into multiple classification. As a result, to the 1500-hour threshold, an alternative methodology to divide data into three distinct groups based on Lifespan and its correlation with other attributes.

The K-Means clustering algorithm was selected as an unsupervised learning algorithm methodology to separate the data into relevant clusters. Using the Elbow Method, which is a technique used to identify the best number of clusters (Cui, 2020), where the k=3 emerged as the most suitable choice because the WCSS displayed considerable decline from k=1 to k=3, which then stabilized. This finding indicates that separating this data into three cluster might be more effective to represent the variation as opposed to binary classification.

Moreover, the binary classification was constructed based on the 1500-hour threshold to meet the early requirement. nonetheless, the triple group classification with K-Means clustering affords a more comprehensive view. This clustering methodology will allow to capture diverse groups that may display different patterns.

4.2 Methodology

Logistic regression was selected because of its simple but efficient when solving binary classification which is observed in (Kirasich et al., 2018) research. Furthermore, it is explained that this model demonstrates a robust performance when classifying between dependent and independent variables. It is a popular model because of its ability to operate and interpret efficiently and formulate the best baseline model to compare. The ANNs were chosen for similar reasons in Part 3, which details to capturing underlying complex non-linear correlations. ANNs are recognized for their adequate managing of layered dimensional data and have a greater generalization performance, making it the ideal model for complicated datasets.

For preprocessing the dataset will contain both numerical and categorical attributes. Similar to Part 3, StandardScaler and One Hot Encoding is used, where StandardScaler will standardize the scale of numerical data to mitigate the risk of any single feature that would dominate the learning process, whereas One Hot Encoding is applied to categorical data into binary representation. The data was split into training, validation and testing sets to ensure that the model is tested using the data that is not seen during training.

The GridSearchCV is used to find the optimal hyperparameter for both models. For logistic regression, both regularisation parameter (C) solver type was experimented on. Similar to Part 3, for ANN hidden layer size, activation functions, learning rate and max iter were optimized to maximise the learning capabilities of the model. Furthermore, cross validation is used to ensure the models ability to generalize effectively and not overfit the training set.

4.3 Evaluation

The evaluation metrics chosen were accuracy, precision, recall and F1-score for the classification task. The accuracy provided a broad overview of the model's performance, whereas precision and recall were critical in dealing with the important imbalances in classes. The F1 score combines both precision and recall identifying where both false positives and false negatives can affect the classification.

For binary classification utilizing the 1500-threshold, the ANN scored a 93% accuracy with a F1 score of 0.87 for instances classified as class 1. The logistic regression only scored a 73% in accuracy with significantly lower recall for class 1 which indicated that the efficiency of detecting defectives parts diminished.

For multi-class classification using the K-Means clustering, both models demonstrated good performance however the ANN was again much better than the logistic regression. The ANN had scored a high F1 score for all classes which is extremely helpful in the production process.

When tuning hyperparameters of Logistic regression, various values of the C parameter and solver types (liblinear, saga, lbfgs) show no major discrepancies. In the case of ANN multiple hyperparameters were chosen like hidden layer ((50,), (100,), (100,50)) and activation functions (ReLu, tanh). This allowed to identify the better model was the hidden layer of (100,), the activation function of ReLu which provided the highest accuracy during the cross-validation process.

Table 7 Logistic Regression Classification Report

	Precision	Recall	F1 Score	Support
0	0.78	0.89	0.83	74
1	0.47	0.27	0.34	26
Accuracy			0.73	100
Macro Avg	0.62	0.58	0.59	100
Weighted Avg	0.70	0.73	0.70	100

Table 8 ANN Classification Report

	Precision	Recall	F1 Score	Support
0	0.97	0.93	0.95	74

1	0.83	0.92	0.87	26
Accuracy			0.93	100
Macro Avg	0.90	0.93	0.91	100
Weighted Avg	0.93	0.93	0.93	100

The confusion matrix for logistic regression revealed a significant number of false positives (8) and false negatives (19), which suggest that it was difficult to differentiate between classes. The confusion matrix for ANN illustrated a superior performance with a mere 5 false positives and 2 false negatives. This shows that the ANN model avoided misclassifying parts as faulty when they were not. Therefore, making the ANN the most recommendable model.

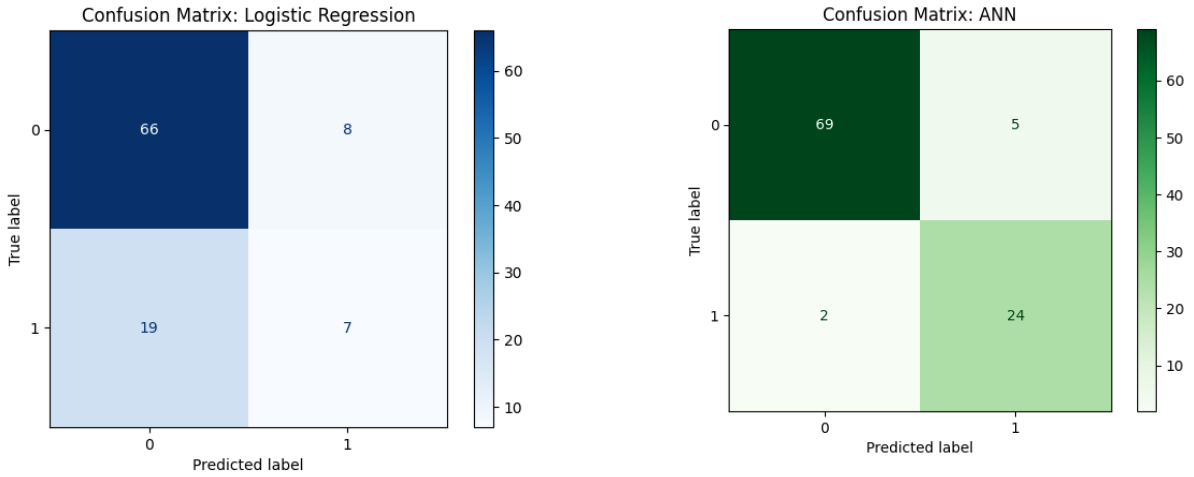


Figure 3 Confusion Matrices

4.4 Critical Review

The overall methodology that was used to interpret this dataset was well suited. The use of feature crafting K=Means clustering to group the data into three clusters, which provided a better understanding of the data compared to generic binary classification. This might be the reason this specific method contributed to the superior performance of the ANN model.

To improve the current results, a possible new model of Gradient Boosting could be implemented, as it could manage both the numerical and categorical data which was causing a complex non-linear relationship. Furthermore, the use of Hierarchical Clustering could have revealed meaning groups which were created by analysing the common relationship among every data point at various levels which allows for deeper classification.

5. Conclusions

The experiment revealed that Regression and Binary Classification models were able to effectively predict the metal parts durability based on the Lifespan. In Part 2, it was identified that there was a significant relationship between certain features and Lifespan which helped when selecting the models. The experiments help enhance understanding on how models should be selected based on evaluating their performance through accuracy and generalization.

The models chosen for regression were the Linear Regression and the Artificial Neural Network (ANNs) were used to predict the Lifespan. In this the ANN yielded as the better performing model with an R^2 of 0.92 which shows that the ANN modelling was more successful in interpreting the complex non-linear data relationships. For the binary classification task regarding Lifespan exceeding 1500-hours threshold, the ANN again performed much better than the Logistic Regression yielding an accuracy of 93%, whereas the Logistic Regression only yielded a 73% accuracy even though it faced challenges because of the class imbalance.

In the context of the business, where they have to predict all the metal parts that are usable being they have a Lifespan greater than 1500-hours is extremely crucial. In this context the Binary Classification methodology fits the company's requirements perfectly. Due to how well the ANN performed, couples with high accuracy and capacity to manage complex relationships, it makes it a more appropriate model for implementation. Therefore, to recommend the best model for both use cases would be the ANNs model, which is justified by its high accuracy and R^2 while also managing to manage any feature complexity as well as providing a practical application in classifying whether any metal parts are usable or not.

6. References

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